Arduino Nano - Setup and Code Upload

(Step by Step)

This file provides a complete, easy, and practical guide for Arduino Nano (ATmega328) — from installing the IDE/drivers to uploading your first code, using the Serial Monitor, and troubleshooting common issues.

1. Quick Introduction / Requirements

Hardware:

- Arduino Nano (typically with Mini-USB connector)
- USB Mini-B cable (data supported)
- PC (Windows / macOS / Linux)

Software:

- Arduino IDE

2. Install Arduino IDE

Windows

1. Download Arduino IDE from the official Arduino website.
2. Run the installer and select "Install USB driver" if prompted.
3. Connect Nano with USB cable — Windows should detect a new COM port (check in Device Manager → Ports (COM & LPT)).

macOS

1. Download Arduino IDE and move it to Applications.
2. If your Nano uses CH340, install the CH340 driver (sometimes macOS security settings require manual "Allow").

Linux (Ubuntu/Debian)

- Extract official Arduino IDE archive and run the arduino executable, or install via:
- sudo apt install arduino
- If you get a USB permission error, add your user to the dialout group:
- sudo usermod -a -G dialout $USER

Then log out and log in again.

3. USB Driver (if COM Port Not Detected)

- Many clone Nanos use **CH340/CH341** USB-to-serial chips → install CH340 driver.
- FTDI-based Nanos may need FTDI drivers (often included in OS).
- Make sure you are using a proper data cable (not a charge-only cable).

4. Arduino IDE Settings (First Time)

1. Open Arduino IDE.
2. Go to Tools → Board → Arduino Nano.
3. Under Tools → Processor, select either **ATmega328P** or **ATmega328P (Old Bootloader)**. If upload fails, switch between them (many clones use old bootloader).
4. Go to Tools → Port and select the correct COM port (Windows: COM3/COM4, Linux: /dev/ttyUSB0 or /dev/ttyACM0).
5. For Serial Monitor, select the same baud rate as in your code (usually 9600 or 115200).

5. First Program — Blink

1. Open File → Examples → 01.Basics → Blink.
2. Example code:

```
void setup() {
 pinMode(LED_BUILTIN, OUTPUT);
}
void loop() {
 digitalWrite(LED_BUILTIN, HIGH);
 delay(1000);
 digitalWrite(LED_BUILTIN, LOW);
 delay(1000);
}
```

3. Verify Board/Port settings, then click **Upload** (right arrow button).
4. Onboard LED should start blinking every second.

6. Using Serial Monitor

- Add Serial.begin(9600); inside setup().

- Use Serial.print() or Serial.println() to send data.
- Open Tools → Serial Monitor and select the correct baud rate.

Example:

```
void setup() {
  Serial.begin(9600);
}
void loop() {
  Serial.println("Hello from Ecruxbot");
  delay(1000);
}
```

7. Installing Libraries

- Open Sketch → Include Library → Manage Libraries...
- Search for the required library (e.g., DHT sensor, Adafruit_Sensor) and click Install.
- Use it in code with #include <LibraryName.h>.

8. Common Troubleshooting

- **Upload Error:** Try switching between “ATmega328P” and “ATmega328P (Old Bootloader)”.
- **Board not detected:** Check USB cable, install correct CH340/FTDI driver.
- **Permission error (Linux):** Add user to dialout group.
- **Code not running:** Check wiring, ensure correct COM port, verify power supply.

Now your Arduino Nano is ready for coding, uploading, and experimenting!